KAIZEN MATHS

Lesson starter

Equation of a Circle

Intersections

Name _____

S1

Find the intersection points of the circle and line:

$(x - 1)^2 + (y + 3)^2 = 25$

$y = -3$

S2

Find the intersection points of the two circles:

$(x + 2)^2 + y^2 = 25$

$(x - 2)^2 + y^2 = 9$

S3

Find the intersection points of the circle and line:

$(x + 1)^2 + (y + 3)^2 = 25$

$y = -3$

S4

Find the intersection points of the two circles:

$(x + 1)^2 + y^2 = 25$

$(x - 3)^2 + y^2 = 9$

Show your working

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GEO-9D18-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Equation of a Circle

Intersections

Name _____

E1

Find the intersection points of the circle and line:

$x^2 + (y + 1)^2 = 25$

$y = -5$

E2

Find the intersection points of the two circles:

$(x + 3)^2 + y^2 = 25$

$(x - 1)^2 + y^2 = 17$

E3

Find the intersection points of the circle and line:

$(x - 3)^2 + (y - 2)^2 = 25$

$y = -1$

KAIZEN MATHS

Exit ticket

Equation of a Circle

Intersections

Name _____

E1

Find the intersection points of the circle and line:

$x^2 + (y + 1)^2 = 25$

$y = -5$

E2

Find the intersection points of the two circles:

$(x + 3)^2 + y^2 = 25$

$(x - 1)^2 + y^2 = 17$

E3

Find the intersection points of the circle and line:

$(x - 3)^2 + (y - 2)^2 = 25$

$y = -1$

GEO-9D18-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Equation of a Circle

Intersections

Name _____

E1

Find the intersection points of the circle and line:

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Find the intersection points of the two circles:

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$(x - 1)^2 + y^2 = 17$

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Find the intersection points of the circle and line:

$(x - 3)^2 + (y - 2)^2 = 25$

$y = -1$

GEO-9D18-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Equation of a Circle

Intersections

Name _____

E1

Find the intersection points of the circle and line:

$x^2 + (y + 1)^2 = 25$

$y = -5$

E2

Find the intersection points of the two circles:

$(x + 3)^2 + y^2 = 25$

$(x - 1)^2 + y^2 = 17$

E3

Find the intersection points of the circle and line:

$(x - 3)^2 + (y - 2)^2 = 25$

$y = -1$

GEO-9D18-L3 · E

Show your working

Teacher answers & working

Equation of a Circle · Intersections

GEO-9D18-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find the intersection points of the circle and line:
$(x - 1)^2 + (y + 3)^2 = 25$
$y = -3$

Answer:

(-4, -3) and (6, -3)

Working:

Substitute y = -3 into the circle equation.

$(x - 1)^2 + (-3 + 3)^2 = 25$

$(x - 1)^2 + 0 = 25$

$(x - 1)^2 = 25$

$x = -4$ or $x = 6$.

The intersection points are (-4, -3) and (6, -3).

S2

Find the intersection points of the two circles:
$(x + 2)^2 + y^2 = 25$
$(x - 2)^2 + y^2 = 9$

Answer:

(2, 3) and (2, -3)

Working:

Expand or subtract the equations to eliminate y^2.

$((x - 2)^2 + y^2 = 9) - ((x + 2)^2 + y^2 = 25)$ gives a linear equation in x.

Solving the linear equation gives x = 2.

Substitute x = 2 into the first circle.

$(2 + 2)^2 + y^2 = 25$

$y^2 = 9$, so $y = 3$ or $y = -3$.

The intersection points are (2, 3) and (2, -3).

Teacher answers & working

Equation of a Circle · Intersections

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S3

Find the intersection points of the circle and line:
$(x + 1)^2 + (y + 3)^2 = 25$
$y = -3$

Answer:

(-6, -3) and (4, -3)

Working:

Substitute y = -3 into the circle equation.

$(x + 1)^2 + (-3 + 3)^2 = 25$

$(x + 1)^2 + 0 = 25$

$(x + 1)^2 = 25$

$x = -6$ or $x = 4$.

The intersection points are (-6, -3) and (4, -3).

S4

Find the intersection points of the two circles:
$(x + 1)^2 + y^2 = 25$
$(x - 3)^2 + y^2 = 9$

Answer:

(3, 3) and (3, -3)

Working:

Expand or subtract the equations to eliminate y^2.

$((x - 3)^2 + y^2 = 9) - ((x + 1)^2 + y^2 = 25)$ gives a linear equation in x.

Solving the linear equation gives x = 3.

Substitute x = 3 into the first circle.

$(3 + 1)^2 + y^2 = 25$

$y^2 = 9$, so $y = 3$ or $y = -3$.

The intersection points are (3, 3) and (3, -3).

Teacher answers & working

Equation of a Circle · Intersections

GEO-9D18-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Find the intersection points of the circle and line:

$x^2 + (y + 1)^2 = 25$

$y = -5$

Answer:

(-3, -5) and (3, -5)

Working:

Substitute y = -5 into the circle equation.

$x^2 + (-5 + 1)^2 = 25$

$x^2 + 16 = 25$

$x^2 = 9$

$x = -3$ or $x = 3$.

The intersection points are (-3, -5) and (3, -5).

E2

Find the intersection points of the two circles:

$(x + 3)^2 + y^2 = 25$

$(x - 1)^2 + y^2 = 17$

Answer:

(0, -4) and (0, 4)

Working:

Expand or subtract the equations to eliminate y^2.

$((x - 1)^2 + y^2 = 17) - ((x + 3)^2 + y^2 = 25)$ gives a linear equation in x.

Solving the linear equation gives x = 0.

Substitute x = 0 into the first circle.

$(0 + 3)^2 + y^2 = 25$

$y^2 = 16$, so $y = -4$ or $y = 4$.

The intersection points are (0, -4) and (0, 4).

Teacher answers & working

Equation of a Circle · Intersections

GEO-9D18-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Find the intersection points of the circle and line:

$(x - 3)^2 + (y - 2)^2 = 25$

$y = -1$

Answer:

(-1, -1) and (7, -1)

Working:

Substitute y = -1 into the circle equation.

$(x - 3)^2 + (-1 - 2)^2 = 25$

$(x - 3)^2 + 9 = 25$

$(x - 3)^2 = 16$

$x = -1$ or $x = 7$.

The intersection points are (-1, -1) and (7, -1).